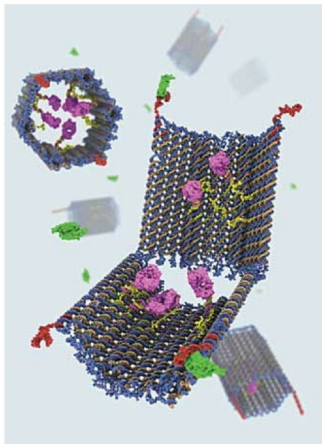


Nanorobotics: Are You Game To Pop A Robot?

Recent technological developments indicate that doctors might one day start prescribing nanorobots as pills for serious ailments!



Cell-targeting DNA nano-robots bearing antibody-fragment payloads (Image created by Campbell Strong, Shawn Douglas, & Gael McGill using Molecular Maya & cadnano)

JANANI GOPALAKRISHNAN VIKRAM

Researchers at Harvard University have recently developed nanoscale robots using DNA, to seek and destroy cancer cells. These tiny clam-shaped bots hold a dose of medicine inside them. They probe the body for cancerous cells and release the medicine only at such locations, leaving healthy cells untouched. This way, the efficacy of the drug is very

high, as the treatment is focused on affected areas.

Dr Shawn Douglas and his team used what they call 'DNA origami' to build the nanorobot. Like in the art of origami, they have been playing with various shapes and intelligent software for a long while. Their software is completely aware of how DNA base-pairs bind together, as well as the shapes they create. When you enter a shape, the software returns a list of DNA strands that can be mixed together to create it.

After experimenting for a while with such a setup, Dr Douglas's team chose a

clam-like caged shape so that the bot could hold the drug carefully inside it. They fitted the cage with two locks—each being a strand of DNA (aptamer) designed to recognise a specific molecule corresponding to affected cells. When the aptamer and the target molecule connect, the DNA strand opens and the drug is released.

In an experiment, they released millions of such nanorobots into a mixture of healthy and cancerous hu-

man blood cells. After three days, they found that around half of the leukaemia cells had been destroyed, leaving healthy cells untouched. By adding more drugs and locks, the nanorobots can be designed to vanquish almost all leukaemia cells in a patient's body.

"Having two locks means that a nanorobot is better able to distinguish between cells," said Dr Douglas, in a recent press report. "Two different signals have to be present to open it, increasing its specificity."

Dr Douglas also hopes that the cancer-targeting nanorobots will steer clear of other rapidly dividing cells, such as those in the gut and at hair follicles, which often suffer collateral damage during chemotherapy.

Medicine of the future

Elsewhere, researchers are exploring how a combination of nanorobots and stem cells can be used to regenerate dead cells in the brain, retina and other vital parts of the body. Stem cells are basic building blocks that are capable of producing new cells to regenerate damaged organs or nerves. Simply injecting stem cells into the body does not help much, as the efficacy of the treatment is too low. So researchers are trying to attach the stem cells to nanofibres or nanorobots and then inject the combination precisely at the damaged location.

A 'scaffold' of nanofibres would support the stem cells in the process of regenerating the damaged parts. Various materials may be used for these nanofibre scaffolds. Polymers and biological molecules such as peptides AND proteins are the main options today.